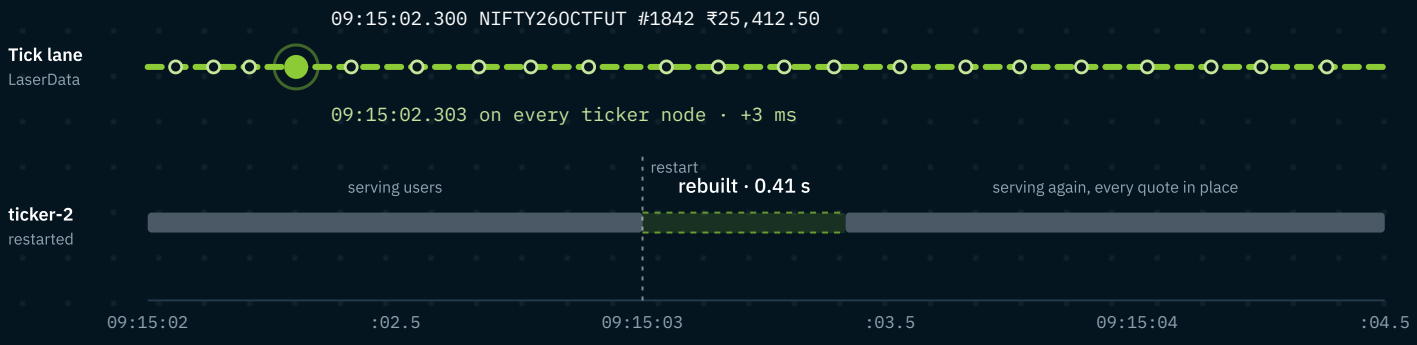
laserdata

Prepared for **Zerodha**

Real-time market ticks for Kite at Zerodha, on LaserData.

Every exchange tick reaches **every Kite ticker node in milliseconds**, in order per instrument, and a node that restarts rebuilds every quote from the log. It runs beside NATS, so nothing else has to move.

One contract, one ticker restart,
as the event log sees it.
Now multiply by every instrument.



09:15:02.300	tick	NIFTY26OCTFUT · #1842 · ₹25,412.50 · NFO
09:15:02.303	ticker.applied	ticker-1, ticker-2 · +3 ms · pushed to every watcher
09:15:03.000	node.restart	ticker-2 · replays the last 75 s of the log
09:15:03.410	quotes.rebuilt	60,000 of 60,000 · even an option idle since 09:14:41

From the creators of APACHE IGGY

15×
faster P99 latency
than Apache Kafka

4×
higher throughput
than Apache Kafka

60–70%
lower infrastructure cost
compute, storage and network

Apache Iggy vs Apache Kafka, LaserData testing. Results vary by workload.

What you shared	The ticker streams **tens of millions of market data packets a second** to over a million users, on NATS after custom TCP, ZeroMQ and Nanomsg. "Latency is THE metric." (Zerodha Tech Blog)
Where it fits	**The tick lane:** exchange ticks keyed by instrument token, one partition per segment, read by every ticker node and kept for rebuilds and replay. NATS, Kafka and Redis stay.
Next	**AI agents** act on the same lane, with session, context, memory and replay from Laser SDK.
Underneath	**One durable event log.** LaserData, powered by Apache Iggy.

Try it in 15 minutes. Let the numbers decide.

A ready-made kit for Zerodha: **60,000 instruments at 20,000 ticks/s** in Kite's quote packet, ticker nodes for 50,000 users each, a 10× burst, a node restart, an order-postback burst next door and a NATS shadow bridge. It prints your lane's p99 every second.

Open the Zerodha kit and write-up →

Then a 2–4 week evaluation on one mirrored tick subject, alongside NATS. **No advantage, no migration.**

kranti@laserdata.com laserdata.com

© 2026 LaserData, Inc. Apache, Apache Iggy and the Iggy logo are trademarks of The Apache Software Foundation; their use does not imply endorsement by the ASF. The Zerodha and Kite names identify the intended reader only; no affiliation is implied. Facts are from Kailash Nadh's posts on the Zerodha Tech Blog (2020, 2021) and the CNCF case study on Zerodha (2020). Instruments, prices, sequence numbers and timestamps are illustrative.